MOHAK SHARMA
25K-3002

OOP CLASSTASKS 04

Task Four:

You've been tasked with designing the core components of a social media app similar to Instagram. The app will allow users to create profiles, post content, interact with posts (e.g., liking, commenting), and view their feed. There are different types of users, each with specific functionalities and access levels.

Tasks:

User Class Design:

- Design a base class User to represent common attributes and functionalities shared by all users, including username, email, and password.
- Implement user verification and password encryption to enhance security.

[Choose a suitable encryption algorithm (e.g., bcrypt, Argon2) for securely hashing passwords.]

Derived User Classes:

- Create derived classes for different types of users: RegularUser, and BusinessUser.
- Each derived class should inherit from the User class and provide specialized behavior based on the user's role and access level.
  - RegularUser Class:
    - Limited Posting: Regular users can post a maximum of 5 posts. Implement logic to enforce this limit.
    - Interactions: Regular users can like posts, comment on posts, and view their feed.
      - The RegularUser class maintains an array feed to store pointers to Post objects.
      - The addToFeed() method adds a post to the feed if there is space available.
      - The viewFeed() method displays the posts in the feed by iterating over the array and calling the display() method of each Post object.
      - Note: max feed size is 10; static const int MAX_FEED_SIZE = 10;
  - BusinessUser Class:
    - Post Promotion: Business users can promote their posts to reach a larger audience. Implement a method to promote posts.
      - User Validation: Ensure that only BusinessUser objects can invoke the promotePost() method.

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```
public class User {
    private String username;
    private String email;
    private String password;

    public User(String username, String email, String password) {
        this.username = username;
        this.email = email;
        this.password = password;
    }

    public String getUsername() {
        return username;
    }

    public void setUsername(String username) {
        this.username = username;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }

    public String getPassword() {
        return password;
    }

    public void setPassword(String password) {
        this.password = password;
    }

    public boolean verifyUser(){
        //Yahan sara encrption wagera logic ayega
        return true;
    }
}
```

```
public class RegularUser extends User{
    private final int Limit = 5;
    private Post[] posts = new Post[5];
    private int numberOfposts = 0;

    public RegularUser(String username, String email, String password, int
numberOfposts) {
        super(username,email,password);
        this.numberOfposts = 0;
    }

    public int getLimit() {
        return Limit;
    }
```

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```
    public Post[] getPosts() {
        return posts;
    }

    public void setPosts(Post[] posts) {
        this.posts = posts;
    }

    public int getNumberOfposts() {
        return numberOfposts;
    }

    public void setNumberOfposts(int numberOfposts) {
        this.numberOfposts = numberOfposts;
    }

    public void viewFeed(){
        for (int i = 0 ;i<Limit;i++){
            if (posts[i] != null) {
                posts[i].display();
            }
        }
    }

    public void addtoFeed(Post post){
        for (int i = 0; i < Limit; i++) {
            posts[numberOfposts] = new
Post("Hello","sijdbv.jpg",false,false,"");
            numberOfposts++;
        }
    }

    public void likePost(Post post){
        if(post!=null)
            post.setLiked(true);
        else
            System.out.println("Post does not Exist");
    }
}
```

```
public class BusinessUser extends User{

    private boolean canPromote;

    public BusinessUser(String username, String email, String
password,boolean canPromote) {
        super(username,email,password);
        this.canPromote = canPromote;
    }

    public void promotePost(Post post){
        System.out.println("Post is being promoted");
    }
}
```

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```
public class Post{
    private String caption;
    private String imgLink;
    private boolean liked;
    private boolean viewed;
    private String Comment;

    public Post(){}

    public Post(String caption, String imgLink, boolean liked, boolean
viewed, String comment) {
        this.caption = caption;
        this.imgLink = imgLink;
        this.liked = liked;
        this.viewed = viewed;
        Comment = comment;
    }

    public String getCaption() {
        return caption;
    }

    public String getImgLink() {
        return imgLink;
    }

    public boolean isLiked() {
        return liked;
    }

    public boolean isViewed() {
        return viewed;
    }

    public String getComment() {
        return Comment;
    }

    public void setCaption(String caption) {
        this.caption = caption;
    }

    public void setImgLink(String imgLink) {
        this.imgLink = imgLink;
    }

    public void setLiked(boolean liked) {
        this.liked = liked;
    }

    public void setViewed(boolean viewed) {
        this.viewed = viewed;
    }

    public void setComment(String comment) {
        Comment = comment;
    }

    public void display(){
```

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```
        System.out.println("Post Img: "+this.getImgLink()+"\nPost
Caption"+this.getCaption());
    }
}
```

```
public class Main {


    public static void main(String[] args) {

        RegularUser u1 = new
RegularUser("mohak","Mohak@mail.com","12345mohkak",0);

        Post posts = new Post();
        u1.addtoFeed(posts);

        u1.viewFeed();

        BusinessUser b1 = new BusinessUser("MOHAK
BUSINESSMAN","Mohak@mail.com","12345mohkak",true);

        b1.promotePost(posts);


        //        RegularUser u2 = new RegularUser
    }
}
```

OUTPUT:

Post Img: sijdbv.jpg

Post CaptionHello

Post Img: sijdbv.jpg

Post CaptionHello

Post Img: sijdbv.jpg

Post CaptionHello

Post Img: sijdbv.jpg

Post CaptionHello

Post Img: sijdbv.jpg

Post CaptionHello

Post is being promoted

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Process finished with exit code 0